

Topic 1. THE EVOLUTION OF SCIENCE AND TECHNOLOGY.

Necessity is the mother of invention



There is a phrase I learned in college called "Having healthy disregard for the impossible". That is a really cool phrase. You should try to do things that most people would not (Larry Page, co-founder of Google, Inc.)

DISCOVERIES AND INVENTIONS

INVENTIONS TIMELINE.

Work in pairs. Look at the following inventions. Decide when they were invented/discovered. Choose the three most important inventions and say why you have selected these three.

Group 1

Wireless telegraphy, printing press, gunpowder, hypodermic syringe, dynamite, blue jeans, terrestrial globe, telegraph, garbage bags.

Group 2

Compact disc, corkscrew, airplane, ball-point pen, transistor, milk carton, toothpaste tube, lightbulb, cinematograph.

Group 3

Laptop computer, roleplaying games, smart card (or chip card), computer mouse, pull-tab can, compass, satellites, violin, beginning of navigation.

Group 4

Catapult, hourglass, hot air balloon, lock and key, glasses (BE spectacles), telescope, television, screw, railroad.

Group 5

Safe, Celsius scale, typewriter, discovery of bronze, stethoscope, pencil, phonograph, wheel, Metric system.

Group 6

Introduction of agriculture, photocopier, taming of fire, locomotive, atomic bomb, colour television, fingerprinting, cave paintings, discovery of penicillin.

Group 7

Toothbrush, diving suit, adhesive tape, 1st Model T Ford, whiskey, zeppelin, telephone, accordion, iPod.

Group 8

Lock and key, electrical cell, plastic, tea bag, flush toilet, bow and arrow, paper, X-rays, washing machine.

Group 9

Anaesthetics, abacus, electric kettle, matches, postage stamp, iPad, radar, internal combustion engine, elevator.

Group 10

Hearing aid, spinning wheel, thermometer, Braille, MP3, manned space flight, condom, lawnmower, glass.

DISCUSSION PRACTICE.

Work in groups of 3-4. Look at ALL the inventions and answer the question: What invention(s) has had the greatest impact on our lives? Why?



15 minutes. Report on the results.

WATCH THE FOLLOWING VIDEO

World-Changing Accidental Scientific Discoveries (10 min)

<https://www.youtube.com/watch?v=LwZML38Mokc&t=66s>

While watching the video pay attention to the use of tenses:

Alexander Fleming **left** a Petri dish and **went** on a vacation. When he **came** back, he **found** a mysterious mold growing that broke down the bacteria.

(Here the verbs in Simple Past are used to show **succession of past actions.**)

Wilhelm Roentgen **was investigating** cathode rays when he **realized** that a new kind of ray passed through a black cardboard tube.

(Here the verb in Past Progressive **was investigating** is used to talk about an action **in progress** in the past, a process, whereas the verb in Simple Past **realized** denotes **a fact** in the past, which interrupted/occurred at the same time as the past progressive action.)

TASK:

1. Describe each of the accidental scientific discoveries in 2 or 3 sentences paying attention to the correct use of tenses. The following notes will help you:

Pacemaker (1958): Wilson Greatbatch – designs a circuit to help record heart sounds – mistakenly grabs the wrong resistor and plugs in circuit – the circuit pulses for 1.8 milliseconds – stops for 1 second and repeats.

Saccharin (1878): Constantin Fahlberg – works on coal tar derivatives – notices sweet taste on his hands – realizes it is from the benzoic sulfamide compound

Safety Glass (1903): Edouard Benedictus – puts away glass flask – flask contains liquid plastic – drops flask on floor – flask shatters but does not break – thin plastic film holds pieces together

Vulcanized Rubber (1839): Charles Goodyear – works with raw rubber powder containing sulphur – brushes powder off hands – powder falls onto hot stove – forms tough elastic substance

Microwave Oven (1945): Percy Spencer – works on active radar set – candy bar melts in his pocket – high-power microwave beams

Dynamite (1867): Alfred Nobel – works in his lab on nitroglycerine to make it safer – vial of nitroglycerine slips out of his fingers – falls to the floor – no explosion – falls onto a pile of sawdust which absorbs the liquid – later mixes with silica

Anesthesia (1842): Crawford Long – notices people under the influence of ether do not feel pain – waits for 7 years – Dr. Green performs the first public demonstration of ether as an inhalable anesthetic

X-rays (1896): Wilhelm Roentgen – investigates cathode rays - projects light from cathode ray generator onto wall – sees outline of bones of own hand on wall

Nuclear Fission (1934): Enrico Fermi – tries to bombard uranium with neutrons – believes he creates new elements – split the nucleus of a heavy element – nuclear power and nuclear bomb

Penicillin (1928): Alexander Fleming – grows bacteria in lab – goes on vacation - penicillium mold in Petri dish kills bacteria

2. Look at the following definitions

Discover – to find something that was hidden or that people did not know about before.

Invent – to make, design, or produce something new for the first time.

Invent vs. discover: you invent something that did not exist before (e.g. machine, method) but you discover something that existed before but was not known (e.g. place, thing, fact).

TOPIC RELATED VOCABULARY

A. Use the following words to complete the sentences with the correct idioms: ***think, move, broke, toyed***

1. The engineers _____ with the idea for some time before they finally decided on the latest features of the product's design.
2. The new invention was so radical that it totally _____ the mold.
3. When you are facing unexpected problems, it helps to _____ outside the box.
4. In a fast-paced industry like IT, you really have to _____ with the times and follow the latest developments.

B. Choose the right word:

Throughout history mankind has created 1) _____ inventions. Among them we find everything from machines like the steam engine which 2) _____ the world of transport, to medical 3) _____ such as cochlear implants. These have advanced the human race but there have been a few 4) _____ ones along the way that haven't quite come up to scratch. For instance, air-conditioned shoes aren't exactly a 5) _____ of genius.

1. **A** idealistic **B** innovative **C** imaginary **D** informative
2. **A** intended **B** innovated **C** produced **D** revolutionized
3. **A** breakthroughs **B** priorities **C** findings **D** resources
4. **A** repulsive **B** ill-conceived **C** ravishing **D** delightful
5. **A** hint **B** sliver **C** speck **D** stroke

C. Complete the sentences with words derived from the words in bold.

1. Inventors always have to _____ come up with new ideas and products.

(continue)

2. One of the great innovations of recent years is _____ energy. **(new)**

3. The _____ of the department is underway with the installation of new computers. **(modern)**

D. Idioms related to technology.

Fill in **light years**, **rocket science**, **plug**, **wires crossed**, **wavelength**. Explain the meaning of the idioms.

1. They **got their** _____ and showed up at different times.
2. I don't know why she is finding it so hard to write a product recall; **it's not** _____.
3. The team works well on the project as the team members **are on the same** _____.
4. The funding has been cut and they **are pulling the** _____ on this research project.

READING

Work in groups of 3. Each student reads one part of the text. Retell to the other students in the group what your part is about. Check understanding of the vocabulary in bold. Choose 2-3 other words that are new/useful for you and share within your group.

3 medical innovations fueled by COVID-19 that will outlast the pandemic

A number of technologies and tools got a chance to prove themselves for the first time in the context of COVID-19. Three researchers working in gene-based vaccines, wearable diagnostics and drug discovery explain how their work rose to the challenge of the pandemic, and their hopes that each technology is now poised to continue making big changes in medicine.

Genetic vaccines

Deborah Fuller, Professor of Microbiology, University of Washington

Thirty years ago, researchers for the first time injected mice with genes from a foreign pathogen to produce an immune response. Like many new discoveries, these first gene-

based vaccines had their ups and downs. Early mRNA vaccines were hard to store and didn't produce the right type of immunity. DNA vaccines were more stable but weren't efficient at getting into the cell's nucleus, so they failed to produce sufficient immunity.

Researchers slowly overcame the problems of stability, getting the genetic instructions where they needed to be and making them **induce** more effective immune responses. By 2019, academic labs and biotechnology companies all over the world had dozens of promising mRNA and DNA vaccines for infectious diseases, as well as for cancer in development or in phase 1 and phase 2 human clinical trials.

When COVID-19 struck, mRNA vaccines in particular were ready to be put to a real-world test. The 94% efficacy of the mRNA vaccines surpassed health officials' highest expectations.

DNA and mRNA vaccines offer huge advantages over traditional types of vaccines, since they use only genetic code from a pathogen – rather than the entire virus or bacteria. Traditional vaccines take months, if not years, to develop. In contrast, once scientists get the genetic sequence of a new pathogen, they can design a DNA or mRNA vaccine in days, identify a lead candidate for clinical trials within weeks and have millions of doses manufactured within months. This is basically what happened with the coronavirus.

Gene-based vaccines also produce precise and effective immune responses. They stimulate not only antibodies that block an infection, but also a strong T cell response that can clear an infection if one occurs. This makes these vaccines better able to respond to mutations, and it also means they could be capable of **eliminating** chronic infections or cancerous cells.

The hopes that gene-based vaccines could one day provide a vaccine for malaria or HIV, cure cancer, replace less effective traditional vaccines or be ready to stop the next pandemic before it gets started are no longer far-fetched. Indeed, many DNA and mRNA vaccines against a wide range of infectious diseases, for treatment of chronic infections and for cancer are already in advanced stages and clinical trials. As someone who has been working on these vaccines for decades, I believe their proven effectiveness against COVID-19 will **usher in a new era of vaccinology** with genetic vaccines **at the forefront**.



Smartwatches and other wearable technologies allow users to capture more continuous health data than ever before. Pixabay

Wearable tech and early illness detection

Albert H. Titus, Professor of Biomedical Engineering, University at Buffalo

During the pandemic, researchers have taken full advantage of the **proliferation** of smartwatches, smart rings and other wearable health and wellness technology. These devices can measure a person's temperature, heart rate, level of activity and other **biometrics**. With this information, researchers have been able to track and detect COVID-19 infections even before people notice they have any symptoms.

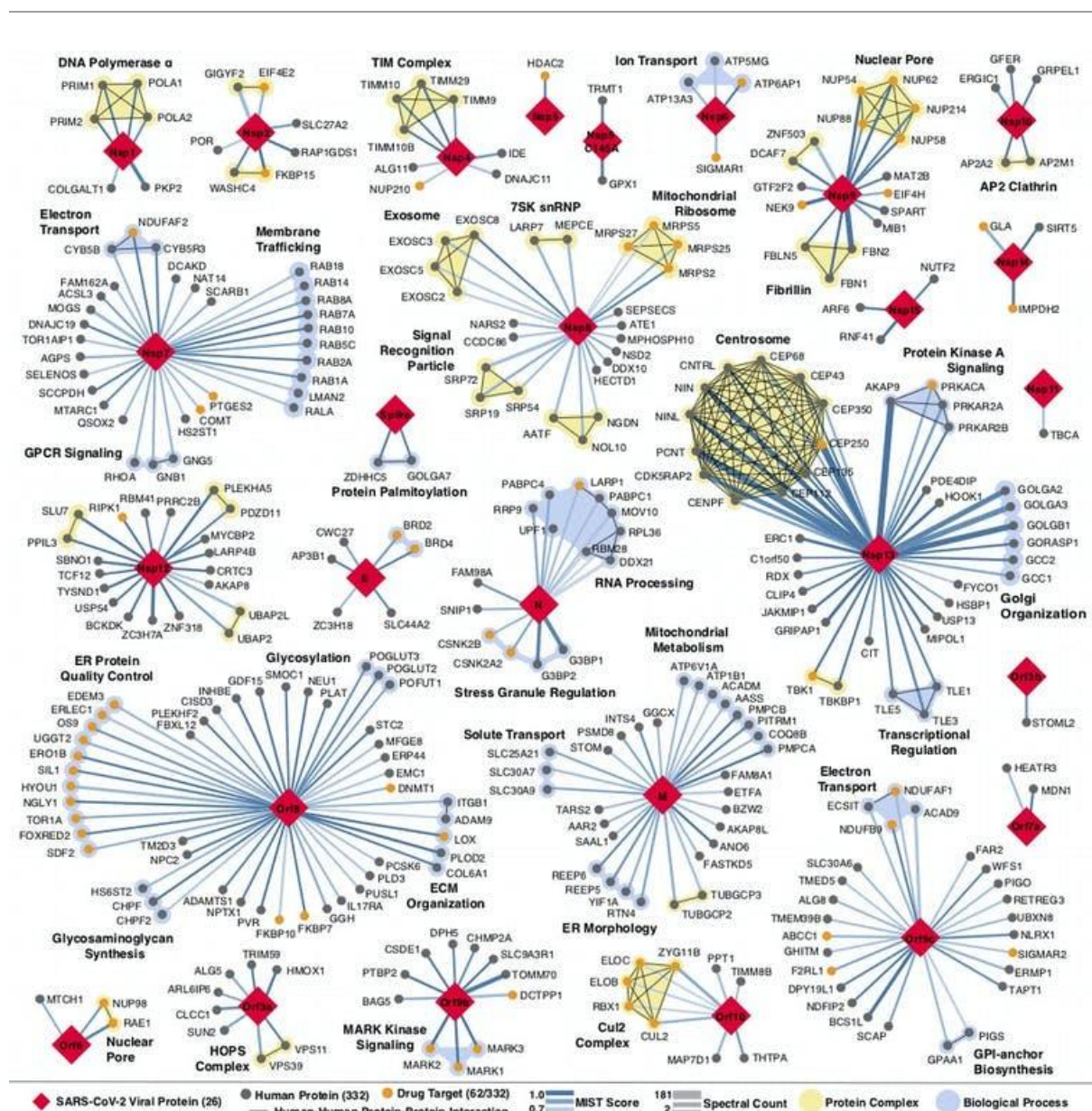
As wearable usage and adoption grew in recent years, researchers began studying the ability of these **devices to monitor disease**. However, although real-time data collection was possible, previous work had focused primarily on chronic diseases.

But the pandemic both **served as a lens to focus** many researchers in the field of **health wearables** and offered them an **unprecedented opportunity** to study real-time infectious disease detection. The number of people potentially affected by a single disease – COVID-19 – at one time gave researchers a large population to draw from and to test hypotheses on. Combined with the fact that more people than ever are using wearables with health monitoring functions and that these devices collect lots of useful data, researchers were able to try to diagnose a disease solely using data from wearables – an experiment they could only dream of before.

Wearables can detect symptoms of COVID-19 or other illnesses before symptoms are noticeable. While they have proved to be capable of detecting sickness early, the symptoms wearables detect are not unique to COVID-19. These symptoms can be predictive of a number of potential illnesses or other health changes, and it is much harder to say what illness a person has versus simply saying they are sick with something.

Moving into the post-pandemic world, it's likely that more people will **incorporate wearables into their lives** and that the devices will only improve. I expect the knowledge

researchers have gained during the pandemic on how to use wearables to monitor health will form a starting point for how to handle future outbreaks – not just of viral pandemics, but potentially of other events such as food poisoning outbreaks and seasonal flu episodes. But since wearable tech is concentrated within pockets of **affluent** and younger populations, the research community and society as a whole must simultaneously address the **disparities** that exist.



Every place that a coronavirus protein interacts with a human protein is a potential druggable site. QBI Coronavirus Research Group, CC BY-ND

A new way to discover drugs

Nevan Krogan, Professor of Cellular Molecular Pharmacology and Director of the Quantitative Biosciences Institute, University of California, San Francisco

Proteins are the molecular machines that make your cells function. When proteins **malfunction** or are **hijacked by a pathogen**, you often get disease. Most drugs work by **disrupting** the action of one or several of these malfunctioning or hijacked proteins. So a logical way to look for new drugs to treat a specific disease is to study individual genes and proteins that are directly affected by that disease. For example, researchers know that the BRCA gene – a gene that protects your DNA from being damaged – is closely related to the development of breast and ovarian cancer. So a lot of work has focused on finding drugs that affect the function of the BRCA protein.

However, single proteins working in isolation are usually not solely responsible for disease. Genes and the proteins they encode are part of complicated networks – the BRCA protein interacts with tens to hundreds of other proteins that help it perform its cellular functions. My colleagues and I are part of a small but growing field of researchers who study these connections and interactions among proteins– what we call protein networks.

For a few years now, my colleagues and I have been exploring the potential of these networks to find more ways drugs could **ameliorate disease**. When the coronavirus pandemic hit, we knew we had to try this approach and see if it could be used to rapidly find a treatment for this **emerging threat**. We immediately started mapping the extensive network of human proteins that SARS-CoV-2 hijacks so it can replicate.

Once we built this map, we pinpointed human proteins in the network that drugs could easily target. We found 69 compounds that influence the proteins in the coronavirus network. 29 of them are already FDA-approved treatments for other illnesses. On Jan. 25 we published a paper showing that one of the drugs, Aplidin (Plitidepsin), currently being used to treat cancer, is 27.5 times more potent than remdesivir in treating COVID-19, including one of the new variants. The drug has been approved for phase 3 clinical trials in 12 countries as a treatment for the new coronavirus.

But this idea of mapping the protein interactions of diseases to look for novel drug targets doesn't apply just to the coronavirus. We have now used this approach on other pathogens as well as other diseases including cancer, neurodegenerative and psychiatric disorders.

These maps are allowing us to connect the dots among many seemingly disparate aspects of single diseases and discover new ways drugs could treat them. We hope this approach will allow us and researchers in other areas of medicine to discover new therapeutic strategies and also see whether any old drugs **might be repurposed to treat other conditions**.

This article originally appeared in 'The Conversation' in March 2021. It is reproduced here under its CC license. <https://theconversation.com/3-medical-innovations-fueled-by-covid-19-that-will-outlast-the-pandemic-156464>.

TERMINOLOGY: Connection
to attach an attachment

*To **attach** something to an object means to join it or fasten it to the object.*

The plate is **attached** to the wall by four screws.

The vacuum cleaner has special **attachments** for narrow spaces.

to connect a connection

*When you **connect** a pipe, wire, etc. to something you join the end of it to the end of the other thing.*

Connect the hose pipe to the tap.

The house is **connected** to the electric grid by underground cables.

There must be a loose **connection**.

 There are serious questions **connected** with radioactive waste disposal.

to link	a link
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*If one place or object is **linked** to another or if something **links** them, there is a physical connection between them so that you can travel between them or communicate between them.*

The television camera had been **linked** to a computer.

 Is there a **link** between smoking and cancer?

to mount

*If you **mount** an object in a particular place, you fix it there firmly on something that will support it.*

Machine guns had been **mounted** on the roof of the police station.

 In IT **to mount** is to make a group of files in a file system structure accessible to a user or user group. In some usages, it means to make a device physically accessible.

The location in VFS that the newly mounted medium was registered is called **mount point**; when the mounting process is completed, the user can access files and directories on the medium from there. An opposite process of mounting is called **unmounting**.

to pivot	a pivot / pivotal
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*If something **pivots**, it turns on a central point.*

Now bring your body forward with a straight back, **pivoting** from the hips.

Success or failure **pivoted** on a single exam.

My days revolved around two **pivots**: the university and the pub.

1941 was the **pivotal** year of the war.

to secure	to be secure
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*If you **secure** something you fasten it firmly to another object or you tie it together tightly.*

A plastic box was **secured** to the wall by ten screws.

Check that the leads to the battery are in good condition and **secure**.

to support	a support
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*If something **supports** an object, it holds it firmly from below, so that it does not drop downwards.*

The walkway is **supported** on/by girders.

Most large buildings now have steel **supports**.

to suspend

*If something is **suspended** from a high place, it is hanging from there, usually so that it can swing or move about.*

The lamps are suspended from the ceiling by ornate chains.

to wind

*If you **wind** something flexible around something else, you wrap it around it several times.*

The coil is **wound** around an iron core.

to wire

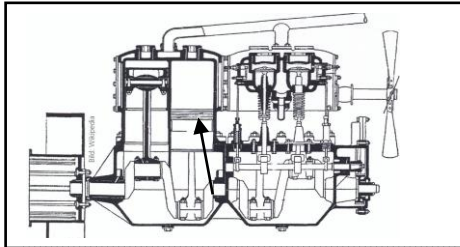
*If you **wire** something or wire it up, you connect it to something else with electrical wires so that electricity or electrical signals can pass between them. If you wire one thing to another, you fasten them together using wire.*

Forty-eight homes were **wired** to the police station.

All the windows in the house are **wired** to discourage burglars from entering.

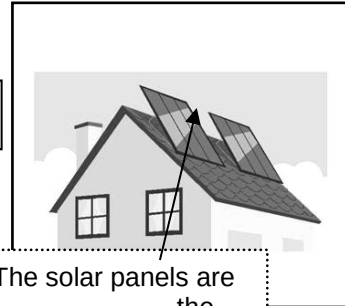
Exercise 1. First look at the words in the box. Then fill the gaps that describe the pictures.

fix to... support by... mount on...suspend from...wind around...
connected to (2x)... fit over...fit into...



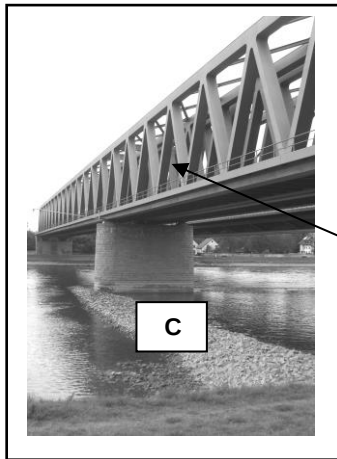
A

The piston is the crankshaft by the connecting rod.



B

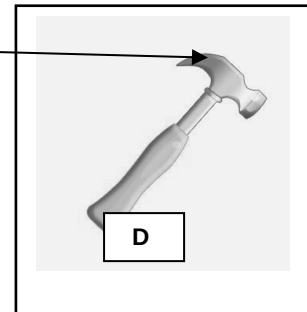
The solar panels are _____ the roof.



C

The head of the hammer is _____ to the handle.
(It cannot move.)

The bridge deck is _____ the pylon.



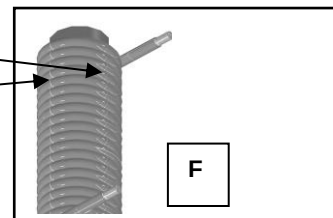
D

A wire is
.....
an iron core.

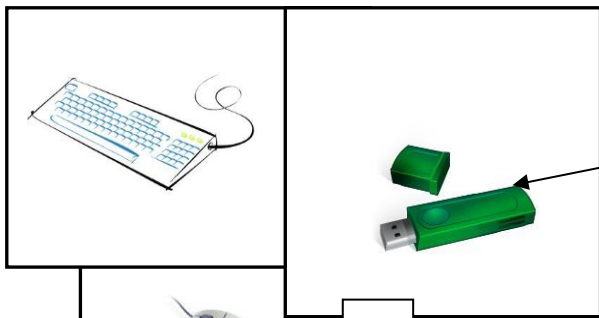


E

The vehicle is
..... a track.



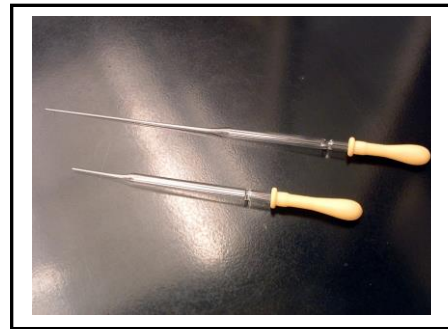
F



These peripherals must be
.....
to the computer.

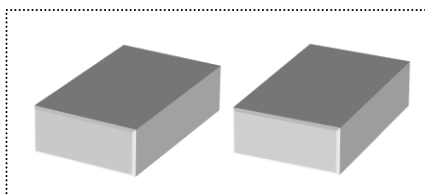


Airbags are
..... the centre of the steering
wheel.

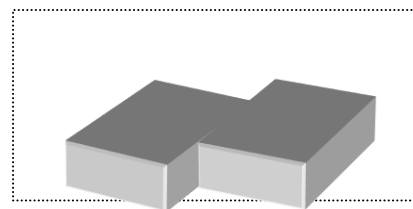


Small rubber bulbs are
.....
..... the top end of the pasteur
pipettes.

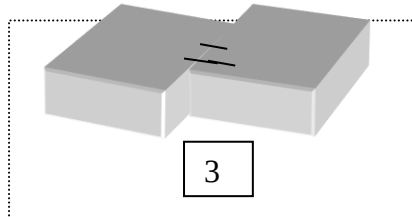
Exercise 2. The following three pictures show two blocks. Which of the following words describe their position relative to each other: *joined* / *adjoining* / *touching* / *separate* / *apart* / *adjacent (to each other)*.



1



2



3

Exercise 3. Complete the following sentences with appropriate words (**nouns, verbs, adjectives, adverbs, and prepositions**).

1. The blackboard is _____ the front wall by eight long screws.
2. The Golden Gate Bridge is _____ by thick steel cables from two tall towers.
3. If the computer is not _____ a power supply, it will not operate.
4. Few people today doubt that there is a _____ asbestos and cancer.
5. When a number of PCs are _____ within an office, we can speak of a LAN.
6. The camera can be pointed in all directions because it _____ the apex of the tripod.
7. The pillars _____ the freeway bridges must be retrofitted to meet the new earthquake standards.
8. A poor _____ between two optical fibers can cause noise on the line.
9. Never open the housing of an electric device which is still _____ to the power supply.
10. _____ one end of the cable to the plug.
11. The Internet _____ millions of users throughout the world.
12. The monitor _____ on its stand so that the user can adjust its position.
13. The printer cable should be firmly _____ to the proper interface on the PC.
14. The various peripheral devices must be _____ the computer.

15. There's a lot of static on the radio. Could there be a loose _____
somewhere?

16. The satellite antenna is _____ on the roof of the house.

FITTED OVER / INTO

17. The cork is _____ the neck of the wine bottle.

18. A leather cover is _____ the rim of the steering wheel.

19. An airbag is _____ the center of the steering wheel.

20. A sensor is _____ the tip of the probe